

András Balogh

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Szeged, Hungary

OBJECTIVE

PhD researcher and former industrial AI developer, specializing in AI security, robustness, interpretability, and adaptive evaluation, seeking PhD-level research roles, internships, or postdoctoral positions in related fields. Published in top-tier venues (ICLR, ICML, AAAI), with hands-on experience in both research and production environments. Proven ability to conduct independent research, lead both industrial and academic AI projects, and collaborate across disciplines.

EDUCATION

Ph.D. in Computer Science

2023 Feb - 2027 Jan (expected)

University of Szeged

Thesis Project: Interpretability and Robustness of Deep Neural Networks

Advisor: Prof. Márk Jelasity

M.Sc. in Computer Science

2021 Feb - 2023 Jan

University of Szeged

Thesis: Disentangling Functional Properties of Neural Network Representations

Thesis Advisor: Prof. Márk Jelasity

GPA: 4.98 / 5.0

B.Sc. in Business Informatics Engineering

2017 Sep - 2021 Feb

University of Szeged

Thesis: Quantitative Analysis of Brain Perfusion SPECT Images

Thesis Advisor: Prof. János Csirik

GPA: 4.73 / 5.0

Miscellaneous

[CentraleSupélec Summer School on AI](#), Paris-Saclay University, 2021

RESEARCH EXPERIENCE

Research Assistant, Ph.D. Student

Dec 2021 - present

Dept. of Artificial Intelligence, University of Szeged

Machine Learning: Robustness, Interpretability, Applications Project

- Similarity, interpretability, and alignment of neural network representations
- Adversarial robustness, out-of-distribution detection and generalization, backdoor attacks

Visiting Researcher

Oct - Nov 2025

Generative AI Lab, Nanyang Technological University, Singapore

Robustness, Uncertainty and Knowledge Augmentation in Generative Models Project

- Uncertainty quantification and RAG-based knowledge augmentation in medical LLMs
- Adaptive and adversarial evaluation of generative world models for autonomous driving

Research Assistant

Mar 2025 - present

HUN-REN-SZTE Research Group on AI

Exploring Novel Approaches to Human-Centered Reliable AI Project

- Mechanistic interpretability of LLMs
- Adaptive and worst-case evaluation of implicit world models in LLMs

Research Assistant

Nov 2018 - Aug 2021

Dept. of Nuclear Medicine, University of Szeged

Image Analysis in Nuclear Medicine Project

- 3D medical image processing and analysis, fully automatic image registration, lesion detection
- Multimodal analysis of functional (SPECT, PET, fMRI) and anatomical (MRI, CT) images

INDUSTRIAL EXPERIENCE

AI Developer and Data Mining Specialist

Jul 2021 - Feb 2025

Bosch Hungary Ltd.

- Created real-time production line monitoring models and software products that operated on high-dimensional time series data under adverse conditions, e.g., distribution shift and data corruption
- Reduced customer claims by 30% and acute maintenance costs by 60% over a three-year period
- Coordinated customer requirements and R&D tasks for a team of 2-3 developers, mentored interns
- Contributed to AI Core projects as part of cross-disciplinary Digitalization and Applied AI Core Teams

Software Engineer

Jun 2018 - Sep 2019

FrontEndArt Ltd.

- Contributed to the development of SourceMeter, a static source code analysis software
- Designed and implemented the storage method and back-end API for human-in-the-loop code quality evaluation

TEACHING EXPERIENCE

Teaching Instructor

Sep 2019 - present

Institute of Informatics, University of Szeged

I am a laboratory instructor, and substitute lecturer (since 2025) of the following courses:

- Machine Learning (English M.Sc. course, 2 semesters)
- Algorithms and Data Structures (English and Hungarian B.Sc. courses, 8 semesters)
- Advanced Algorithms and Data Structures (Hungarian B.Sc. course, 6 semesters)

Teaching Assistant

Sep 2018 - Jun 2019

Institute of Informatics, University of Szeged

I was a teaching assistant and tutor of the following courses:

- Foundations of Programming (Hungarian B.Sc. course, 1 semester)
- Object-oriented Programming (Hungarian B.Sc. course, 1 semester)

HIGHLIGHTED PAPERS (4/7)

- ICLR 2026 **András Balogh** and Márk Jelasity. Verification of the Implicit World Model in a Generative Model via Adversarial Sequences. [\[paper\]](#) [\[code\]](#)
- AAAI 2025 **András Balogh** and Márk Jelasity. How not to Stitch Representations to Measure Similarity: Task Loss Matching versus Direct Matching. [\[paper\]](#) [\[code\]](#)
- ICML 2023 **András Balogh** and Márk Jelasity. On the Functional Similarity of Robust and Non-Robust Neural Representations. [\[paper\]](#) [\[code\]](#)
- CVPR 2026 Linxiang Su and **András Balogh**. ATAC: Augmentation-Based Test-Time Adversarial Correction for CLIP. [\[paper\]](#)

HONORS AND AWARDS

- 2025 University Research Excellence Scholarship
- 2024 University Research Excellence Scholarship
- 2023 1st Place at the National Scientific Students' Associations Conference
New National Excellence Program Scholarship
Nokia Young Scientist Award
- 2022 1st Place at the Scientific Students' Associations Conference & Morgan-Stanley Special Prize
New National Excellence Program Scholarship
National Higher Education Scholarship
- 2021 3rd Place at the National Scientific Students' Associations Conference
National Higher Education Scholarship
- 2020 1st Place at the Scientific Students' Associations Conference & Morgan-Stanley Special Prize

PROFESSIONAL SERVICE

Reviewer

I have been a reviewer at the following conferences:

AAAI (2025-), ICLR (2025-), CVPR (2025-), ECCV (2026-)

Thesis Supervisor

I have been supervising thesis projects since 2023 in ML interpretability, robustness, and medical ML applications.

- 3 B.Sc. theses
- 1 M.Sc. thesis

LANGUAGES

- English (bilingual)
- French (intermediate)
- Hungarian (mother tongue)